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On urban microclimate spatial-temporal dynamics: Evidence from the integration of fixed and wearable sensing and mapping techniques

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ABSTRACT

Urban Heat Island (UHI) is acknowledged to generate harmful consequences on human health, and it is one of the main anthropogenic challenges to face in modern cities. Due to the urban dynamic complexity, a full microclimate decoding is required to design tailored mitigation strategies for reducing heat-related vulnerability. This study proposes a new method to assess intra-urban microclimate variability by combining for the first time two dedicated monitoring systems consisting of fixed and mobile techniques. Data from three fixed weather stations were used to analyze long-term trends, while mobile devices (a vehicle and a wearable) were used in short-term monitoring campaigns conducted in summer and winter to assess and geo-locate microclimate spatial variations. Additionally, data from mobile devices were used as input for Kriging interpolation in the urban area of Florence (Italy) as case study. Mobile monitoring sessions provided high-resolution spatial data, enabling the detection of hyperlocal variations in air temperature. The maximum air temperature amplitudes were verified with the wearable system: 3.3 °C in summer midday and 4.3 °C in winter morning. Physiological Equivalent Temperature (PET) demonstrated to be similar when comparing green areas and their adjacent built-up zone, showing up the microclimate mitigation contribution of greenery in its surrounding. Results also showed that mixing the two data acquisition and varied analysis techniques succeeded in investigating the UHI and the site-specific role of potential mitigation actions. Moreover, mobile dataset was reliable for elaborating maps by interpolating the monitored parameters. Interpolation results demonstrated the possibility of optimizing mobile monitoring campaigns by focusing on targeted streets and times of day since interpolation errors increased by 10% only with properly reduced and simplified input samples. This allowed an enhanced detection of the site-specific granularity, which is important for urban planning and policymaking, adaptation, and risk mitigation actions to overcome the UHI and anthropogenic climate change effects.

1. Introduction

The constant increase in the urban population and the consequent cities expansion (United Nations Human Settlements Programme, 2020) is accompanied by changes in the landscape, from natural surfaces to impervious ones, and by anthropogenic activities growth in these areas (Siddiqui et al., 2021). These alterations affect the urban microclimate, making cities warmer than their surroundings, which is known as the Urban Heat Island (UHI) phenomenon (Santamouris, 2020). The UHI is one of the biggest issues currently faced in urban environments due to its several consequences, like enlarged heat-related health risks, compromised air quality, and higher energy consumption for cooling indoor environments (Buchin et al., 2016; Li et al., 2018; Morakinyo et al., 2018; Zhang, 2020; Paolini et al., 2017. UHI becomes even more challenging under the climate change context considering the projections of more intense, frequent, and longer extreme heat events like heatwaves (Maggiotto et al., 2021). In detail, UHI has already been demonstrated to influence the magnitude and duration of heatwaves (Tan et al., 2010),

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and the combined effect of these two phenomena intensifies the heat stress, implying higher risk and vulnerability for the urban population (Founda and Santamouris, 2017; He et al., 2021).

Given the adverse implications of UHI and human health risks associated with urban overheating (Santamouris, 2020), the implementation of effective mitigation strategies is urgent (Pisello et al., 2013). In particular, Tsoka et al. (2018) have obtained maximum surface and air temperature drops by 9.0 °C and 0.85 °C by increasing the albedo of ground surfaces. Kolokotsa et al. (2018) have found a reduction in the superficial temperature by 10 K by replacing conventional pavements with cool paving stones (higher albedo), and Kyriakodis and Santamouris (2018) reduced air and surface temperatures up to 1.5 °C and 11.5 °C, respectively, using cool asphaltic and concrete pavements. Additionally, it is widely demonstrated that greenery is capable to alleviate urban overheating and reduce heat vulnerability and heat-related mortality (Adnan et al., 2022; Chang and Li, 2014; He, 2019; Mahmoud, 2011; Pantavou et al., 2013; Son et al., 2016). Chàfer et al. (2022) found a 4 °C difference in air temperature between green and non-green areas of Singapore, while Pioppi et al. (2022) demonstrated that small urban parks can reduce air temperature by 3–7 °C in New York City.

Urban environments are characterized by complex morphologies that lead to high heterogeneity and microclimate variations (Pigliautile and Pisello, 2020). Therefore, any application of mitigation strategy requires a specific methodological and experimental approach. Further comprehension of hyperlocal variations in the urban microclimate is needed to propose tailored mitigation actions that account for site-specific characteristics (Pigliautile et al., 2021). Usually, urban microclimate assessments are made by means of satellite images (Halder et al., 2021) or data retrieved from fixed weather stations (Cui et al., 2017). Satellite remote sensing has the advantages of large spatial coverage and repetitive data acquisition (Halder et al., 2021). However, the large coverage compromises retrieving data with a high level of spatial details, and even though images are obtained repeatedly, this process is not continuous. This implies that the time range acquisition for the same picture could be more than one day. On the contrary, fixed weather stations guarantee continuous long-term monitoring allowing the identification of temporal variations of environmental parameters, but they have limitations to evaluating individual exposure on a microclimate level (Bailey et al., 2020) as a huge number of systems is needed to catch spatial variabilities (Pigliautile and Pisello, 2020). Afterwards, the use of mobile monitoring systems is an alternative or complementary solution to retrieve environmental data with the temporal and spatial granularities required to detect intra-urban microclimate variations at strategic times during the course of the year and with relatively low cost compared to high maintenance-needing stations' diffuse network. These systems can equip vehicles (Hart and Sailor, 2009; Kousis et al., 2021; Yadav and Sharma, 2018) or portable devices known as wearables (Chokhachian et al., 2018; Cureau et al., 2022; Pigliautile and Pisello, 2020). Their difference is that the vehicle-based systems are limited to roadways and parking areas, while wearable systems overcome this restriction (Cureau et al., 2022). Differently from remote sensing and fixed weather stations, the use of mobile systems implies short monitoring campaigns due to limitations in systems autonomy and error minimization due to the combination of spatial and temporal data variability. A crucial result is that long-term changes in environmental parameters cannot be identified by this technique. Nevertheless, the huge advantage of using this system is that it monitors the most realistic environment to which people are exposed since they record data within the urban canopy, and, in the case of wearables, according with pedestrians' perspective. Therefore, each technique has its positive and negative aspects, and for this reason, the ideal approach to assessing urban microclimate is the optimal combination of them.

Starting from the aforementioned research and from their further extension, this study proposes a novel methodology to evaluate intra-urban microclimate variability by combining two distinguished environmental data acquisition techniques: fixed and mobile experimental monitoring systems. The main objective was to verify whether and how the granular mobile monitoring data complements the continuous, but less granular, information from fixed-weather stations, for assessing microclimate and heat stress variability. In addition, the study aimed to verify whether mobile monitoring data could be useful as input for spatial interpolation methods. To reach these aims, the analysis of retrieved data focused on the air temperature signal processing by taking advantage of the pros of both the adopted techniques: (i) long-term data history and complete daily profiles in a single point from the fixed weather stations; (ii) high spatial granularity in short-term monitoring from the mobile systems. Two different mobile systems (a vehicle-based system and a wearable device) were used to verify hyperlocal variations in the microclimate within the urban area in summer and winter to access strategic parts of the inner city that are not reached by car. Air temperature values retrieved by the mobile systems were thus corrected to neglect the variation component due to elapsed time. The same corrected data were used to map air temperature spatial variation in the whole area within the monitoring path by means of the Kriging interpolation method. Finally, data retrieved by mobile systems within the urban canopy were used to quantify the heat stress in the urban environment by means of the Physiological Equivalent Temperature (PET), its spatial distribution analysis, and the influence of urban greenery on it. Florence (Italy) was selected as case study representative of Italian big historical cities. The Florentine historical center is characterized by a highly variable urban form, including high and narrow urban canyons in the most central area with surrounding green areas. Moreover, because the city has an enormous flow of tourists, the knowledge of thermal map and air temperature distribution is crucial to identify the potential zones of discomfort.

2. Methods

The high microclimate heterogeneity of urban areas, due to the massive variety of their forms, fabrics, and functions, requires an approach of data gathering and analysis capable of combining multiple scales and differentiated big data sources. The new procedure presented in this research combines two techniques for microclimatic data collection: fixed weather stations network and a variety of interoperating mobile devices. The latter further differs in terms of proposed vectors for data collection, namely a vehicle-based and a wearable system.

2.1. Monitoring campaign

The urban microclimate characterization is provided through both long-term and short-term monitoring. The long-term analysis relies upon five years long data series that were collected by fixed weather stations located in the studied area. The short-term monitoring consisted of a dedicated campaign carried out by mobile monitoring systems. Despite many cities worldwide already presenting a dedicated weather station network made by fixed monitoring setup (sometimes located above the urban canopy or in areas where microclimatic variables are not affected by urban peculiarities like airports), it is not common to acquire data from environmental transects. Since this infrastructure is missing, dedicated campaigns must be carefully designed in time and space to provide meaningful insights into the microclimate heterogeneity of the investigated city. In this research, two different routes were defined accounting for the two different vectors available for data collection: a motorized vehicle and a walking person. Both paths were designed to last around 1.5 h. The pedestrian route was internal to the vehicular one, covering a smaller area due to differences in vector speed. They were assumed to be equal to 1.4 m/s for the walking person and between 5.5 and 8.3 m/s (20–30 km/h) for the vehicle at the design stage. The proposed limitation in vehicle speed is in line with previous tested protocols involving motorized vectors for mobile transects, as reported by Kousis et al. (2021). Furthermore, both routes were circular

to start and conclude environmental data collection in the same area. Therefore, the short-term monitoring campaign consisted of simultaneously repeating the predefined strategic paths with the two mobile systems three times per day in different seasons. The initial time of each monitoring session was established at 8:00 a.m., 12:00 p.m., and 4:30 p.m. local time.

2.2. Experimental setup

In this research, two mobile monitoring systems developed at the University of Perugia were used: a vehicle-based system (Kousis et al., 2021) and a wearable device (Cureau et al., 2022). As explained by Kousis et al. (2021), the monitoring setup on the vehicle comprises three air temperature probes on its top (accuracy: ±0.5 °C) and the compact miniaturized weather station GMX501 which embeds a GPS unit and monitors the following environmental data: air temperature (±0.3 °C), relative humidity (±2%), wind velocity (±3% @40 m/s) and direction (±3 deg @40 m/s), atmospheric pressure (±0.5 hPa @25 °C), and incoming solar radiation (spectral range 300 ÷ 3000 nm). The monitoring system settled on the motorized vehicle retrieved instantaneous values of the listed parameters every 10 s (technical details about the system and its reliability are presented in Kousis et al. (2021)). The wearable monitoring system consists of a set of miniaturized sensors settled on a backpack retrieved and sent to a device directly associated with the same system. In addition to the GPS signal, the system collects the following environmental parameters: air temperature (±1 °C), relative humidity (±3%), atmospheric pressure (±0.25%), wind velocity and direction, and incoming solar radiation (±5%). These parameters are retrieved every 5 s by the system. The wearable system validation is detailed in Cureau et al. (2022).

Furthermore, the urban microclimate assessment combined the information retrieved by the two mobile systems with a five-year-long data series gathered from fixed weather stations located in the area of interest. A detailed description of the identified fixed weather stations is provided in Section 3.

2.3. Data analysis

Long-term data from fixed weather stations and short-term data retrieved in dedicated campaigns through mobile systems were analyzed to provide a comprehensive picture of the microclimate heterogeneity of the city. The data analysis process can be summarized in three phases.

(i) comparison of long-term data series retrieved by fixed weather stations to verify the existence of the UHI phenomenon;
(ii) spatial distribution of air temperature data in the area enclosed by the mobile monitoring paths relying upon data retrieved within the urban canopy by both the mobile systems just after a data series correction in time, accounting for the temporal trend observed by the closest fixed weather station;
(iii) spatial distribution of PET computed within the urban canopy as an index of pedestrian exposure to thermal stress conditions.

The analysis of the UHI was based on a statistical description of the five-year-long database from different fixed weather stations that were characterized according to their distance from the city core and difference in height. Furthermore, monthly profiles of 24 h air temperature trends (minimum, maximum, and average values) were derived for all the stations and compared through the assessment of the Full Width at Half Maximum (FWHM), that is, the range between the two points on a curve where the function reaches 50% of its amplitude.

By zooming into the city, below the urban canopy, the microclimate heterogeneity could be expressed by air temperature spatial variation, as collected by the mobile systems, while accounting for the influence of elapsed time. The time-correction process was faced considering the temporal trend retrieved by the fixed system during the short-term monitoring, as a reference, assuming that this trend was linear. Single temperature observations retrieved by the mobile system at the time i were modified to be comparable with data retrieved by the same system at the time zero of the experimental campaign, i.e.:

$$T_{mi,corrected_fix} = T_{mi} + \frac{T_{f0} - T_{ff}}{f} \bullet i\,[^{\circ}C], 0 < i \leq f$$

where T_{mi} is the retrieved temperature data at the time i by the mobile system, T_{f0} and T_{ff} are temperature values gathered by the fixed station at the beginning and the end of the route, f is the specific short-term monitoring duration.

The same air temperature data collected by the mobile monitoring systems and corrected in time were used to compute the air temperature spatial variation in the monitored area by means of the ordinary Kriging interpolation method. Kriging is a geostatistics method for spatial interpolation based on semi-variograms, which considers the statistical relationship between measured points to determine the parameters in the non-monitored points in a spatial grid (Baykal et al., 2022; Yadav and Sharma, 2018). Even though ordinary Kriging is a simplified method that assumes the mean and variance of the values are constant through the considered area, it has been largely used for air temperature spatial interpolation due to its relatively fast computation and acceptable accuracy, while accounting for the spatial dependence between observations taken nearby. In this view, Yadav and Sharma (2018) achieved a maximum error of 0.5 °C, Rosillon et al. (2024) reported a mean error of 0.71 °C, and Wang et al. (2017) observed a mean error of 1.56 °C, all using ordinary Kriging for interpolating air temperature. In this study, the interpolation procedure was replicated for every monitoring session. First, Kriging was performed using all temperature data for the session (about 1,500 values each). Then, the same procedure was done using one-fifth of these data to verify how using a smaller sample affects the accuracy of the interpolation. The comparison between the model using all the data and that one using the one-fifth sample was made considering their Root Mean Square Error (RMSE). Kriging was done in R version 4.1.2 (R Core Team, 2021) with the RStudio development environment (RStudio Team, 2021), using the package *gstat* (Gräler et al., 2016; Pebesma, 2004).

Finally, the impact of urban microclimate on thermal comfort and potential thermal stress zones were analyzed by computing the PET, defined as "the air temperature at which, in a typical indoor setting (without wind and solar radiation), the heat budget of the human body is balanced with the same core and skin temperature as under the complex outdoor conditions" (Höppe, 1999). PET was determined in the software RayMan (Matzarakis et al., 2007) along the monitoring paths for a standard person (characteristics proposed by the software, i.e., a man, 35 years old, 1.75 m height and 75.0 kg weight, wearing garments equivalent to 0.9 clo and with an internal heat production of 80 W) by considering retrieved data of air temperature, relative humidity, solar radiation, and wind velocity. The standard person used for PET calculation was selected to make the results comparable with the ranges of physiological stress proposed by Matzarakis et al. (1999) and presented

Table 1
Ranges of PET and the corresponding thermal perception and physiological stress, considering internal heat production of 80 W and heat transfer resistance of the clothing equal to 0.9 clo (Matzarakis et al., 1999).

PET (°C)	Thermal perception	Grade of physiological stress
<4	Very cold	Extreme cold stress
4–8	Cold	Strong cold stress
8–13	Cool	Moderate cold stress
13–18	Slightly cool	Slight cold stress
18–23	Comfortable	No thermal stress
23–29	Slightly warm	Slight heat stress
29–35	Warm	Moderate heat stress
35–41	Hot	Strong heat stress
>41	Very hot	Extreme heat stress

in Table 1, which used the same characteristics presented above.

3. The case study

The city center of Florence (Italy) represents the case study. Florence is among the ten largest cities in Italy, with around 368,000 inhabitants according to the Italian National Institute of Statistics for 2021 (Italian National Institute of Statistics, 2021). Moreover, the city center of Florence has a high historical value and is full of monuments and cultural heritage sites as worldwide acknowledged. The historical city center presents diverse urban forms with high and narrow urban canyons in the most central area and variable distribution of green areas. The city is visited by more than ten million tourists every year, especially in the summer season. Therefore, the heat stress and thermal discomfort are more intense exactly during the busiest travel period.

The year-round microclimate conditions in different areas of the city center, which are expected to be affected by the UHI phenomenon, were investigated. In particular, areas of touristic flows were preferred since they are often the most crowded. To this aim, data collected within the city through fixed and portable microclimate monitoring stations – described in the previous section – were considered. In detail, three fixed weather stations located in different zones of the city were used. The first one (hereinafter Station I) is located near the airport of Florence Peretola (Consorzio Lamma, 2022), in the sub-urban area of Sesto Fiorentino, just outside the city (about 10 km distance) at about 40 m a.s.l. (Fig. 1b), and retrieves air temperature data every 60 min. The second one (Station II) is located at the Engineering campus of the University of Florence (Banca dati SIR Regione Toscana, 2022), up to a hill at about 95 m a.s.l. in a rather green area just outside the city center (Fig. 1c), gathering data every 15 min. The third one (Station III) is located inside the city center of Florence (Fondazione Osservatorio Ximeniano di Firenze, 2022), about 51 m a.s.l., near the main square of the city (Fig. 1d), and retrieves air temperature and relative humidity data every 30 min.

Mobile monitoring sessions were used to complement the data from the three fixed weather stations, allowing for evaluating intra-urban variability on environmental parameters. The monitoring paths were designed to cross the city center, passing through areas with different characteristics in terms of geometry, morphology, materials, building density, and expected crowd. In fact, the purpose of short-term mobile transects below the urban canopy is to catch hyperlocal peculiarities that could be generated by differences in forms, fabrics, and functions. Nevertheless, dedicated campaigns need to account for the vector typology of the monitoring system and its characteristics, e.g., data collection frequency rate, system velocity, and restrictions in urban areas. The two monitoring systems followed two different paths that crossed each other in some spots starting and finishing always at the same place (the triangular shape in Fig. 2). The vehicle monitored the area just around the historical city center, passing through residential zones and large green areas. In the end, the vehicle also crossed the historical center (Fig. 2a). The wearable station was focused on monitoring the paths described by typical tourist flows since it went through



Fig. 1. Location of the three fixed weather stations (dots) within the city: a) overview map, b) sub-urban area (Station I), c) uphill green area (Station II), and d) city center (Station III) (source: Google Earth).

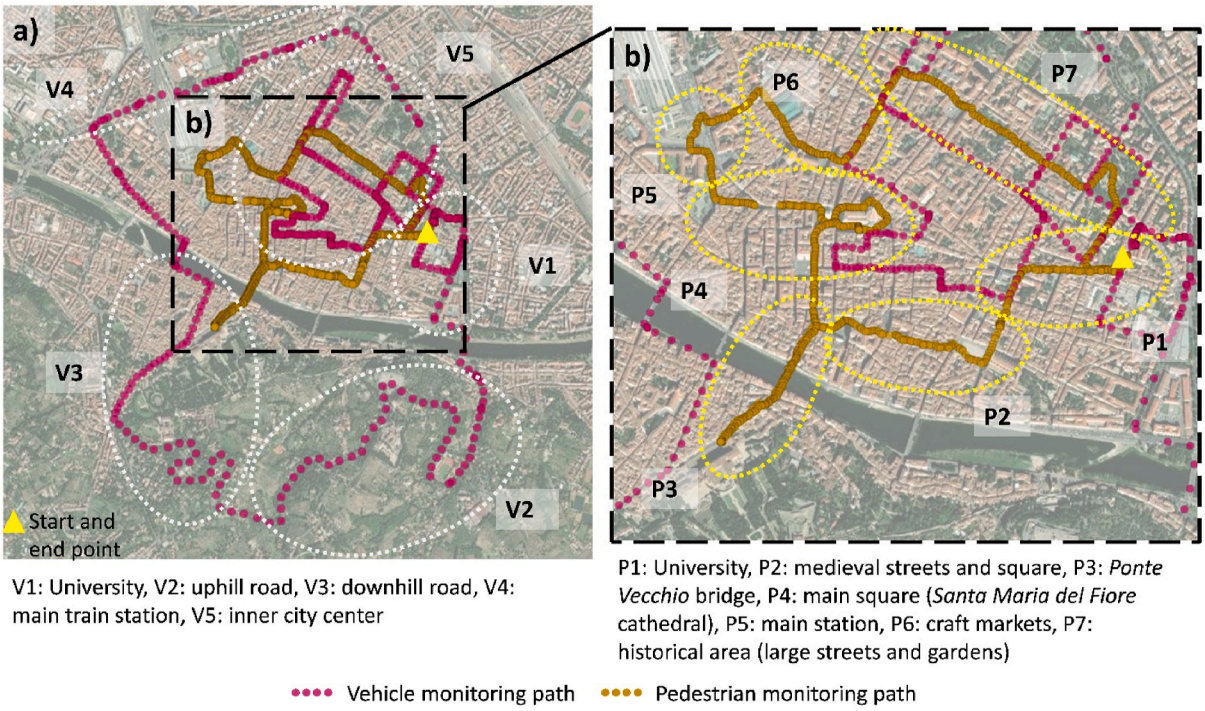


Fig. 2. Monitoring path: a) vehicle, b) pedestrian.

the most important touristic spots and squares of Florence, which are mainly located in the inner-city center (Fig. 2b). The total pedestrian route covered approximately 7.5 km and took about 80 min to complete, while the vehicle route was approximately 25.7 km and required about 100 min.

The vehicle and pedestrian paths (Fig. 2) starting and ending points were at the University (zones V1 and P1 on the maps). Thereafter, the vehicle moved just outside the city center passing by an uphill road (V2), reaching up to 130 m a.s.l. in a mainly green area, and went along a downhill road (V3), going back down to 50 m a.s.l., as the city center. Then, it passed by the main train station of the city (V4) and finally crossed the inner-city center (V5), as shown in Fig. 3a, where it crossed the pedestrian path in various spots. Instead, the pedestrian path, after leaving the University, reached an area characterized by medieval streets and *Piazza Santa Croce* (P2), one of the most important squares in the city. Thereafter, it followed through three of the most important touristic spots: *Piazza della Signoria* (Fig. 3b), *Ponte Vecchio* (P3), and *Piazza San Giovanni*, i.e., the main square of the city where *Santa Maria del Fiore* Cathedral is located (P4). Finally, the path moved near the main train station (P5), *San Lorenzo* (P6), the main historical market of the city, and the historical residential areas with large streets and gardens (P7).

As above-mentioned, two monitoring campaigns were carried out, (i.e., in summer and winter), taking into account warm and cold season conditions, when UHI presents different characteristics. In particular, a sunny day at the end of summer (September 15th, 2021) and a clear sky day at the end of winter (March 11th, 2022) were selected. These days were both typical working days with open schools and routine schedules.

4. Results

4.1. Long-term analysis

This section presents the long-term monitoring results, based on 5-year data (from 2017 to 2021) collected from the three fixed weather stations. Fig. 4 shows the air temperature values recorded by each station every month for five years. The air temperature distribution is similar for all the stations when analyzed month by month except for Station I data from May to December in 2018 and 2021, due to data logging and acquisition problems. In Station II (green area), the air temperature distribution has lower medians than the other two stations. No trends related to air temperature increases were identified over the monitored years. On the other hand, if a 5-years period cannot be a reference for the air temperature increase, it can be significant for the fluctuations linked to the combination between microclimatic parameters and different ground uses, citizens' and tourists' behavior, and building features (changes in materials, installed HVAC systems, architectural features, etc.).

An in-depth comparison between the fixed weather stations was made by determining the minimum, maximum, and average daily profiles of the air temperature for every month. Fig. 5 shows an example of these profiles for the year 2020. The daily profiles are comparable among the stations. It can be noted that at Station I, the temperature begins to rise earlier in the day during the hot season (from May to August). Since this station is located in an open area, it is fully exposed to solar radiation early in the morning and, consequently, it registers an air temperature increase before the other stations. In contrast, the air temperature decreases later at Station III because it is the one in the city center. This is an outcome of the UHI phenomenon, i.e., the heat stored during the day is gradually released at night depending on the thermal capacity of different materials combined with local heat/mass exchange and urban porosity, which causes a lower urban nocturnal cooling rate in urban environments than in non-urban ones. These results are in compliance with Oliveira et al. (2022). The same trends were observed during the 5-year period of available data.

These profiles were compared by means of the FWHM evaluation. This metric was calculated for the average daily profile, and as these curves are not exactly symmetrical, the extreme points that delimit the FWHM were also determined. Table 2 presents an example of these values for the year 2020. Despite the FWHM being similar among the stations, the one located in the city center (Station III) always presented a higher right limit (0.71 h higher on average), which means that its nocturnal cooling rate is lower than the other two stations, confirming



Fig. 3. Monitoring systems: a and b) vehicle-based system, c and d) wearable device.

the UHI effects in the historical city center. Stations I and II had similar right limits, indicating that they have similar cooling rates. Conversely, the left limit was lower at Station I from April to September, which is a result of the incidence of solar radiation that occurs earlier in the morning in its location and accelerates its heating rate during this time of the year. A trend in FWHM increase was not detected.

4.2. Mobile environmental monitoring

The confirmation of the UHI effects in the city center intensifies the demand for focusing on the microclimate analysis below the urban canopy to understand the UHI impacts on people well-being and sustainable environmental conditions. This could be addressed by carrying out environmental transects through mobile monitoring systems as the two simultaneously used in this study (a vehicle-based system and a wearable device). A preliminary analysis of the gathered datasets was carried out in terms of retrieved parameters minimum and maximum values, median, and mean percentage error (calculated considering the median as reference), as reported in Table 3. In general, the errors of the data registered by the vehicle-based system were higher than those collected by the wearable system (except for the winter morning monitoring). This can be a result of the different monitored paths because as the van crossed a longer path covering a larger area, the variation in the environmental parameters registered was generally higher. In addition, wind speed had the highest errors, which occurs as a function of the high variability of this parameter. Indeed, wind speed can be considerably different when comparing two subsequent records, while air temperature and relative humidity vary progressively. The same was observed when analyzing solar radiation, as it can change significantly due to solar availability and shading. In detail, solar radiation varied from 0 W/m^2 (shaded zones) to 769 W/m^2 (area with direct radiation at the midday monitoring) in the vehicle path during winter, and from 0 W/m^2 to 817 W/m^2 in the pedestrian path during the same season and time of the day. In summer, zero radiation was registered in both paths in the shaded areas, while the maximum values were registered during the midday monitoring (731 W/m^2 in the vehicle path and 662 W/m^2 in the pedestrian path). The maximum values are comparable with the ones registered at midday by the fixed weather station located

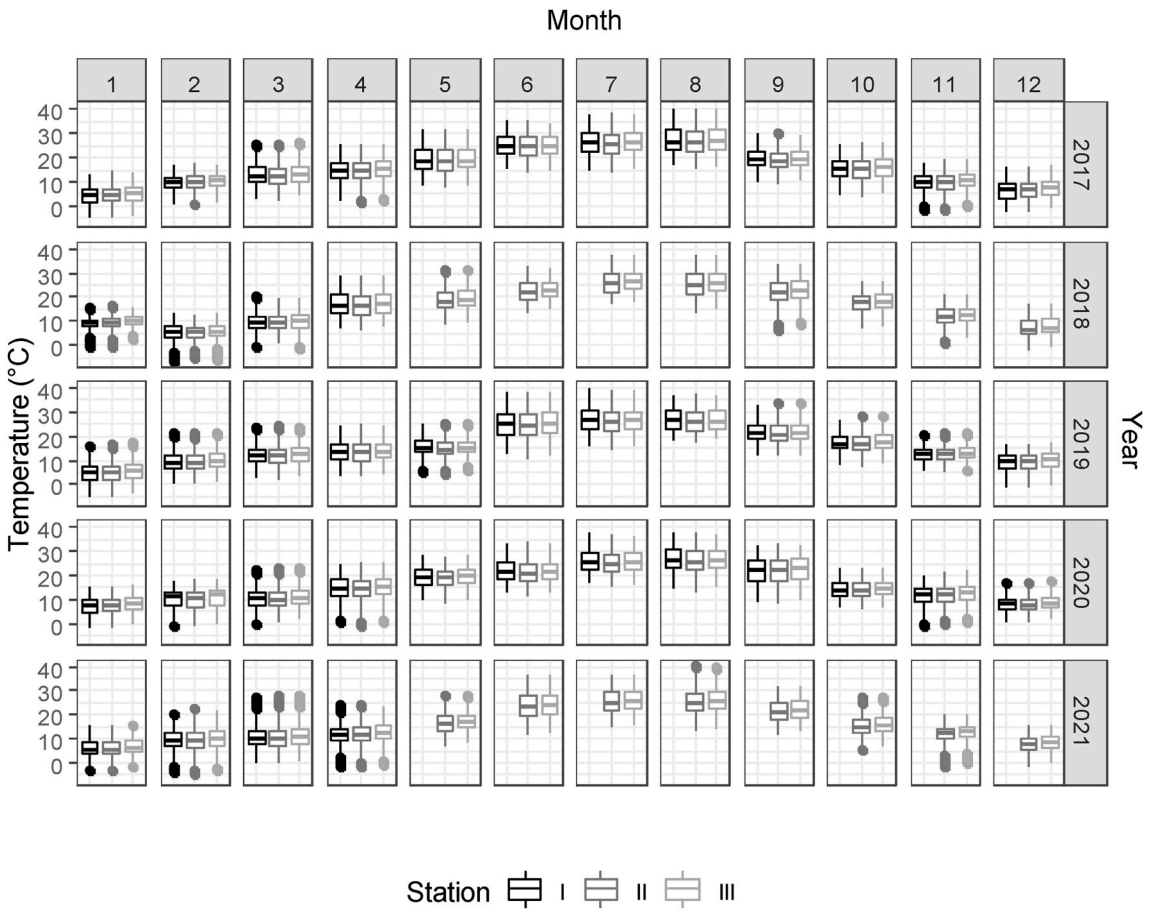


Fig. 4. Air temperature values: assessment from data of the three fixed weather stations over five years.

in the city center (757 W/m^2 in winter and 642 W/m^2 in summer), but the high variability in the parameters identified with the mobile systems enhances the importance of applying this monitoring method to catch the real exposure within urban areas.

Therefore, the hyperlocal intra-urban microclimate variation was investigated through the air temperature registered by the two different mobile systems after correcting these values for the elapsed time. Along the pedestrian path, on the summer day at 8:00 a.m., the highest temperature was registered in the craft markets (*San Lorenzo* historical market zone, P6 in Fig. 2), followed by the main station area (P5), and the lowest, occurred in the main square (*Piazza San Giovanni*, zone P4). Zones P5 and P6 during this period of the day are distinguished by a larger people concentration that can justify the higher temperatures, while P5 is mostly shaded at this time, which delays the air temperature increase. The difference between the maximum and minimum temperatures in this route was 0.9 °C. At 12:00 p.m. and 4:30 p.m., the lowest temperatures were verified in zone P1 and the highest in P7. Even though zone P7 is not touristic, so potentially less crowded, it is characterized by high vehicle traffic, which releases a notable amount of heat. At 4:30 p.m., zones P3, P4, P5, and P6 also presented high air temperatures, connected to the highest crowds at this time. The air temperature amplitude was higher during the midday and afternoon routes: 3.3 °C at 12:00 p.m. and 2.5 °C at 4:30 p.m. Zone P7 was one of the hottest during the three walks, and despite that, it was observed a reduction in air temperature around an urban park (approximately 25,000 m^2) located in this area. Indeed, other studies already demonstrated the cooling effects of greenery in urban environments (Chàfer et al., 2022; Wong et al., 2021).

Relating to the vehicle-based path during summer, the hottest zone was V4 (main train station) at 8:00 a.m., V1 (University) at 12:00 p.m., and both V1 and V4 at 4:30 p.m. These are the areas with the highest concentration of anthropogenic activities due to the existing plenty of facilities and services, releasing significant amount of anthropogenic heat. Some specific spots in the inner city (V5) also had high temperatures at 12:00 p.m. and 4:30 p.m., possibly associated with the crowd. The lowest temperatures were identified in zones V2 and V3. Notably, the section at V2 crosses a green area and, as previously stated, greenery has a significant mitigating effect. Altogether, in the vehicle-based path, the temperature varied by 2.6 °C at 8:00 a.m., 2.7 °C at 12:00 p.m., and 3.0 °C at 4:30 p.m.

In the pedestrian path during winter, the maximum temperature values were registered in P7 at 8:00 a.m. and in P1 and P5 at 12:00 p.m. and 4:30 p.m. P1 and P5 are not touristic areas but have a high flux of people and vehicles due to the university and train station. The lowest temperature values were identified in the *Piazza San Giovanni* (P4) during the three walks. At 8:00 a.m. and 12:00 p.m., P2 and P3 had low temperatures similar to those recorded in P4. These two areas are distinguished by several touristic spaces and a high people concentration (except at 8:00 a.m.). As a consequence, the lower temperatures are strictly connected to the lower exposure to solar radiation. The absolute value of the air temperature variation identified in the pedestrian path was 4.3 °C at 8:00 a.m., 3.2 °C at 12:00 p.m., and 1.8 °C at 4:30 p.m. As in summer, the urban park located in P7 caused an air temperature reduction inside its area and in the immediate surrounding, even at 8:00 a.m., when this zone was the hottest.

Concerning the vehicle-based monitoring in winter, the highest temperatures were verified in V1 at 8:00 a.m. and 12:00 p.m., and in V5 at 4:30 p.m. In contrast, at 8:00 a.m. and midday, V5 was the coolest zone, and at 4:30 p.m., V2 was. Along this path, the temperature value amplitude was around 2.7 °C in all the sessions.

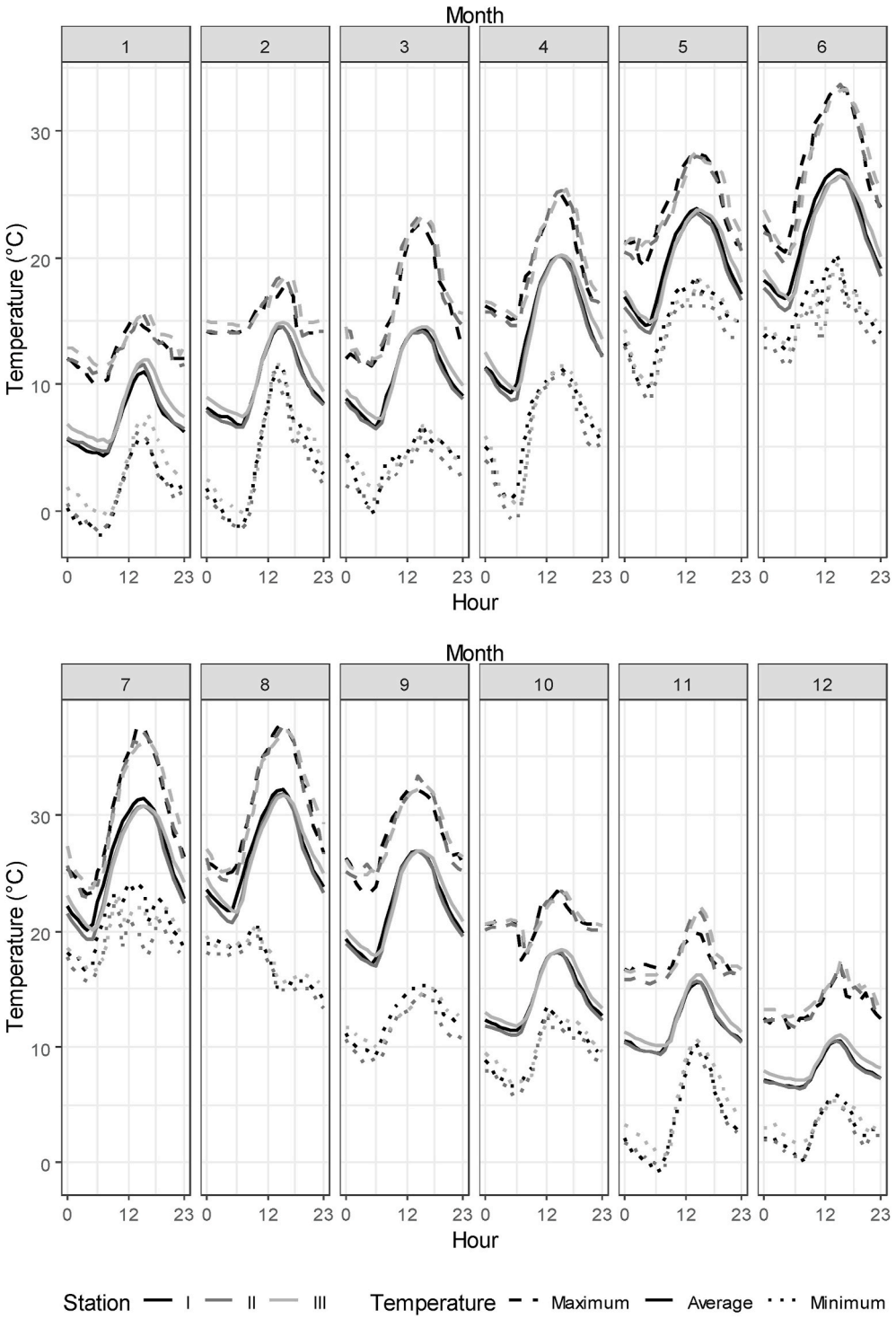


Fig. 5. Minimum, maximum, and average daily profiles of the air temperature registered by the three weather stations in 2020.

Air temperatures from the mobile monitoring systems were compared to Station III, the station located in the area where wearable and vehicle systems collected data. A difference between fixed station and transects was anticipated because while the first is positioned above the urban canyon, the second catches human's real exposure since measurements are taken closer to people. Temperature inside the street canyon (measured by mobile systems) could be higher due to heat accumulation during the daytime and anthropogenic heat release, while fixed stations could also record higher temperatures because of greater solar radiation exposure compared to mobile systems, which traverse shaded and unshaded areas. In winter, both wearable and vehicle registered higher air temperatures than Station III throughout the day, with mean differences of 0.80 °C for the wearable and 0.21 °C for the vehicle. In summer, the wearable had varied patterns: lower temperatures in the morning (mean difference = 0.33 °C) and higher in the midday and afternoon sessions (1.04 °C) compared to Station III. The vehicle recorded lower temperatures in this season (on average, 1.35 °C lower than the fixed station). This higher difference observed between vehicle and Station III could be attributed to the longer path traversed by the vehicle, which includes a large green area, probably cooling the

Table 2
FWHM and its delimiting points (time of the day) for the air temperature average daily profiles during 2020.

Month	Left limit (hour)			Right limit (hour)			FWHM (hour)		
	I	II	III	I	II	III	I	II	III
Jan	10.38	10.40	10.88	19.74	19.48	20.25	9.36	9.08	9.37
Feb	10.16	10.04	10.56	19.49	19.46	20.23	9.33	9.42	9.67
Mar	9.62	9.57	10.18	19.76	19.64	20.39	10.14	10.07	10.21
Apr	9.06	9.20	9.81	19.99	20.03	20.69	10.93	10.83	10.88
May	8.28	8.59	9.21	20.30	20.23	21.12	12.02	11.64	11.91
Jun	8.20	8.48	9.11	20.02	20.26	21.10	11.82	11.78	11.99
Jul	8.44	8.86	9.45	20.32	20.37	21.21	11.88	11.51	11.76
Aug	8.81	9.10	9.69	19.60	19.77	20.47	10.79	10.67	10.78
Sep	8.90	9.12	9.81	19.70	19.51	20.23	10.80	10.39	10.42
Oct	9.64	9.60	10.16	19.32	19.18	19.79	9.68	9.58	9.63
Nov	9.96	9.85	10.49	19.12	19.06	19.54	9.16	9.21	9.05
Dec	10.15	9.98	10.55	19.23	19.05	19.79	9.08	9.07	9.24

Table 3
Minimum, maximum, median and mean percentage error for air temperature (°C), relative humidity (%), and wind speed (m/s) for data retrieved from the mobile monitoring systems.

Route	Parameter	Vehicle-based system				Wearable device			
		Min	Max	Median	Percentage error [%]	Min	Max	Median	Percentage error [%]
Winter morning	Air temperature [°C]	4.9	10.4	7.5	16.1	4.3	12.6	6.5	28.6
	Relative humidity [%]	52.0	73.0	58.0	9.2	38.8	59.4	52.7	9.3
	Wind speed [m/s]	0.1	9.4	1.6	66.9	0.0	4.0	1.4	36.8
Winter midday	Air temperature [°C]	13.2	15.7	14.3	3.5	13.9	17.8	15.1	5.6
	Relative humidity [%]	21.0	36.0	29.0	11.0	19.1	30.6	26.0	10.0
	Wind speed [m/s]	0.1	6.9	1.8	56.3	0.0	5.4	1.9	42.0
Winter afternoon	Air temperature [°C]	12.3	17.3	14.7	4.0	14.0	17.0	15.0	3.4
	Relative humidity [%]	31.0	40.0	36.0	4.4	26.1	33.9	31.2	4.1
	Wind speed [m/s]	0.0	39.0	2.0	54.1	0.0	7.6	1.9	49.6
Summer morning	Air temperature [°C]	18.2	22.1	20.5	4.1	19.3	21.8	20.4	3.4
	Relative humidity [%]	58.0	71.0	63.0	4.8	55.4	61.3	58.8	2.8
	Wind speed [m/s]	0.1	8.3	1.4	67.5	0.0	3.9	1.6	34.4
Summer midday	Air temperature [°C]	23.9	26.8	25.6	2.6	27.0	29.5	28.4	1.8
	Relative humidity [%]	44.0	52.0	48.0	4.0	37.6	44.2	40.8	3.1
	Wind speed [m/s]	0.1	11.0	1.4	66.1	0.0	4.8	1.7	41.6
Summer morning	Air temperature [°C]	25.5	28.7	26.7	1.8	27.5	28.7	28.2	0.8
	Relative humidity [%]	38.0	49.0	43.0	5.1	34.8	43.6	37.5	4.8
	Wind speed [m/s]	0.1	6.9	1.6	57.2	0.0	7.9	1.8	53.3

environment and leading to lower average temperatures. This effect was possibly not observed during winter because the cooling effect of urban greenery is less pronounced in this season (Bartesaghi-Koc et al., 2020; Hamada and Ohta, 2010).

Given the retrieved observations along the monitoring route, an overview of temperature distribution in the whole area enclosed by the same path is given thanks to the Kriging interpolation analysis. Fig. 6 presents the obtained results using all data available for each session for the three monitoring periods on the summer day, as representative example of the method outcomes. Table 4 shows the RMSE of each interpolation model, comparing it when using the entire and one-fifth of the data sample.

In both seasons, the air temperature pattern computed with the entire and one-fifth of the data sample is comparable. However, using all data allowed identifying more extreme values of air temperature. The amplitude of the temperature computed for the non-monitored points was indeed slightly higher when using the entire sample. This was expected as the Kriging method considers the statistical relationship between the measured points: the larger the number of monitored points available, the more probable it will be to distinguish extreme air temperature values. However, this difference did not compromise the reliability of the method when using a smaller sample (about 300 input points) as the RMSE is very similar to the one considering all data (differences lower than 0.1, i.e., about 10%).

Therefore, these results demonstrate that mobile monitoring is reliable for this interpolation method, which is an essential tool for analyzing and scaling up the UHI variation within the same urban area without monitoring the entire zone. Moreover, reducing considerably the number of monitored points did not cause a significant error increase. Therefore, one can achieve similar results with less monitoring effort and, consequently, lower economic investment.

4.3. Physiological Equivalent Temperature (PET)

Thermal comfort along the paths was assessed by means of the evaluation of the PET and the grade of physiological stress proposed by Matzarakis et al. (1999). In this analysis, no time correction was applied to the air temperature since PET includes solar radiation, a parameter that changes over time and cannot be adjusted. As an example, in Fig. 7, the PET variation map along the pedestrian and vehicle routes is shown.

During summer, on both pedestrian and vehicle paths, the zones with the higher air temperature values were not necessarily the ones with higher heat stress. This highlights that solar radiation and wind velocity can increase or attenuate the stress level. In the pedestrian path, the most critical routes were the ones at 12:00 p.m. and 4:30, when most parts of the path were under moderate heat stress (78% at midday and 65% at 4:30 p.m.). At midday, 11% of the route was under strong heat stress. The areas under the worst thermal conditions were P2, P3, P4, and P6, which are the ones with the highest tourist flux. Less thermal stress condition was identified along the vehicle path, with most parts of the route being in the comfortable range (18–23 °C) at 12:00 p.m. and 4:30 p.m.

In winter, most parts of the two routes were in the range of extreme cold stress at 8:00 a.m. For the other periods, the condition was less

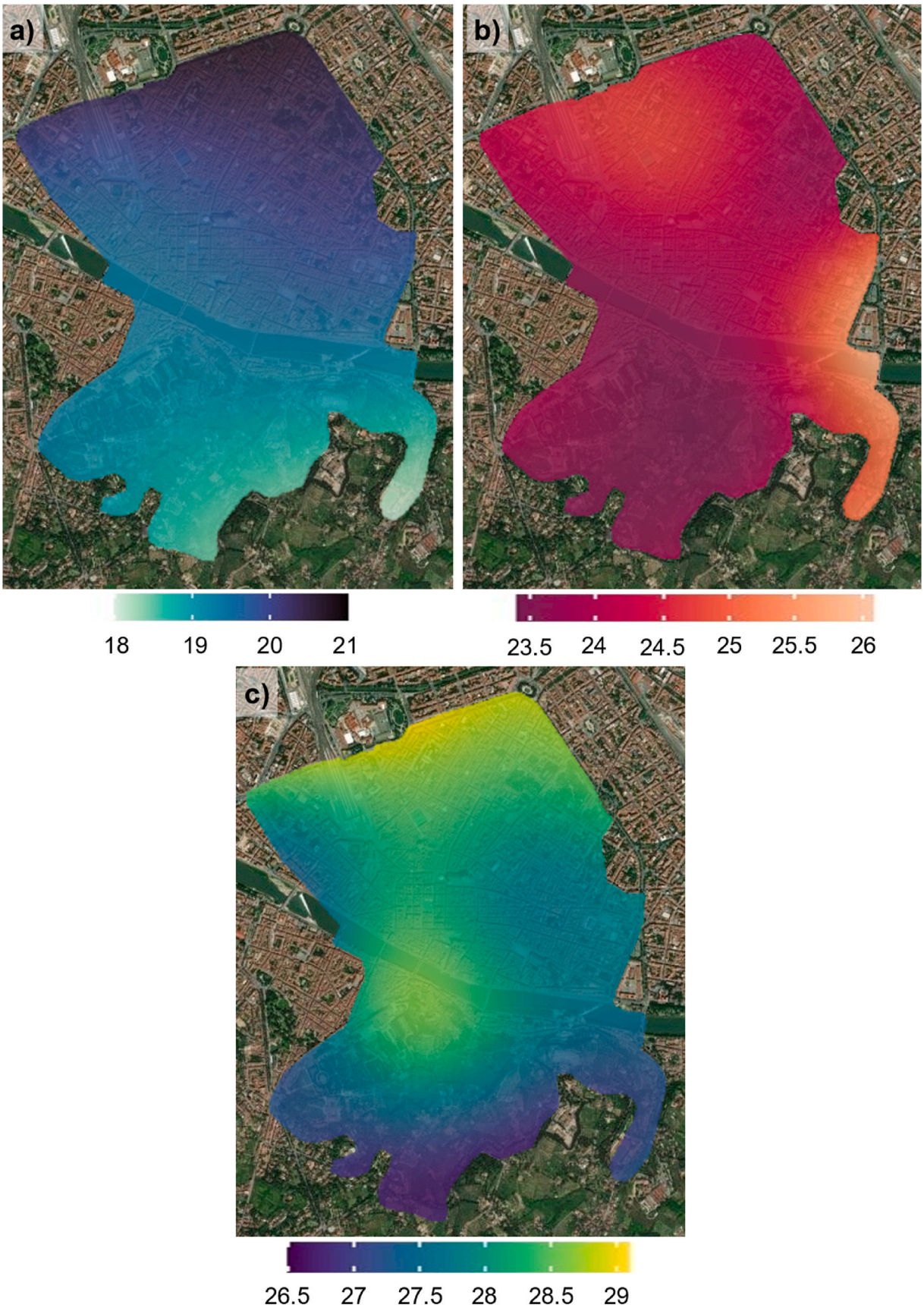


Fig. 6. Air temperature (°C) computed through the Kriging interpolation method on the summer monitoring day at a) 8:00 a.m., b) 12:00 p.m., and c) 4:30 p.m.

Table 4
RMSE resulted from the Kriging interpolation method.

Session	RMSE	
	All data	One-fifth sample
Summer 8:00 a.m.	0.589	0.631
Summer 12:00 p.m.	0.285	0.373
Summer 4:30 p.m.	0.294	0.303
Winter 8:00 a.m.	0.542	0.602
Winter 12:00 p.m.	0.538	0.589
Winter 4:30 p.m.	0.359	0.407

difficult since no extreme thermal stress (below 4 °C or above 41 °C) was identified, but the entire paths were under some type of cold stress (between 4 °C and 18 °C), with exception of the pedestrian route at midday when 10% of it was within the comfort limits. The PET variation in the same route was higher in winter than in summer, with amplitudes higher than 20 °C at 8:00 a.m. and 12:00 p.m.

A more in-depth analysis of the contribution of solar radiation and wind velocity to increase or alleviate heat stress was conducted by comparing a green and a built-up section for each path. In the pedestrian route, the urban park at the end of zone P7 and the street section immediately before the park were compared. Along the vehicle route, the green area crossed in V2 was compared with a densely built street in V3. Short sections (3 min each for the pedestrian path and 6 min for the vehicle one) were selected to guarantee that air temperature and relative humidity had lower variation so that the PET variations were due to changes in solar radiation and wind velocity.

Fig. 8 shows the link between PET, solar radiation, and wind velocity in the compared areas during the summer monitoring at 12:00 p.m. In this figure, it is notable that the built-up area had higher PETs than the green area in the pedestrian route. The higher PETs are connected to higher solar radiation and lower wind velocity in the built-up section, while the green area had faster wind that reduced the PET. In the vehicle path, PET was similar in both areas. Higher solar radiation was registered in the green section, but at the same time, this area reported higher wind velocity, resulting in an alike PET when compared with the built-

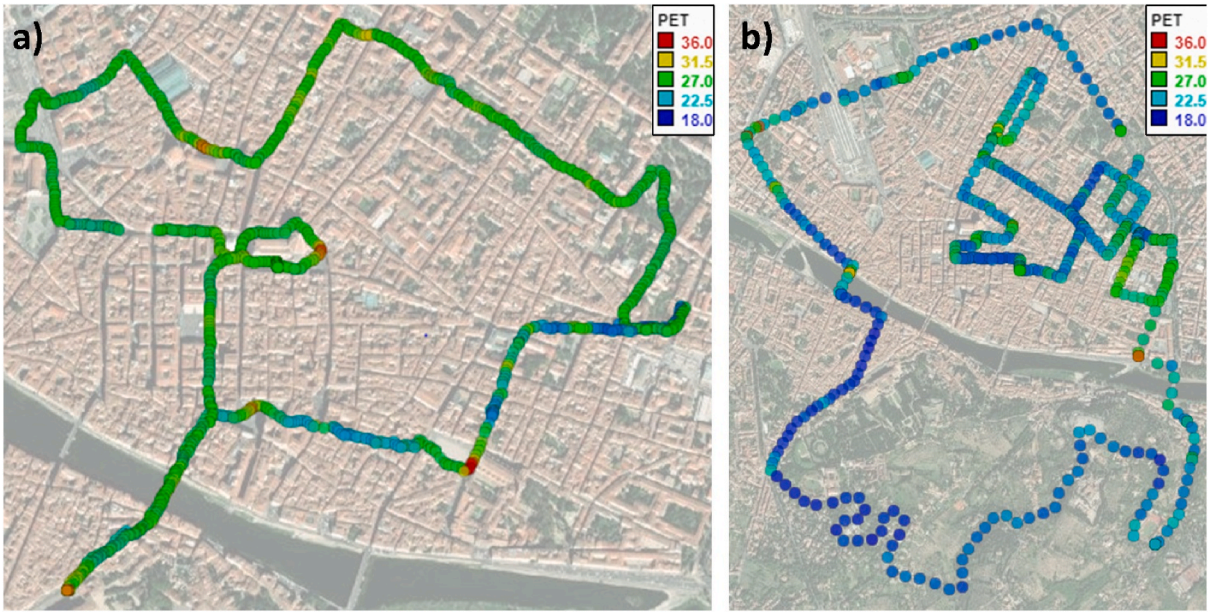


Fig. 7. PET (°C) for the summer monitoring at 12:00 p.m.: a) pedestrian path and b) vehicle path.

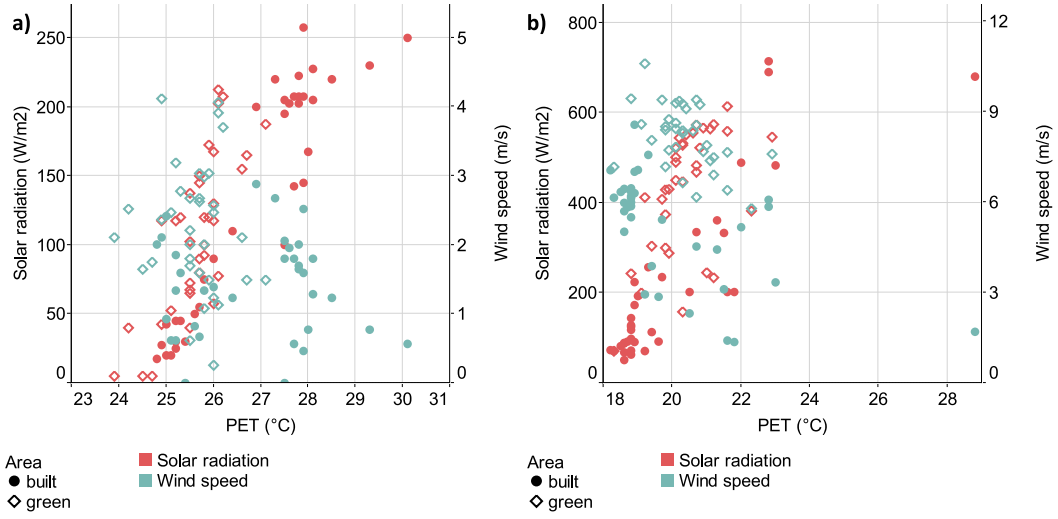


Fig. 8. PET correlations with solar radiation and wind velocity in a green and a built-up area during the summer monitoring at 12:00 p.m.: a) pedestrian path and b) vehicle path.

up area.

The R^2 for these correlations was determined. In this case, a higher R^2 could indicate that PET is more affected by a main parameter under the considered conditions. In summer, at 8:00 a.m. and 12:00 p.m., it was verified a better correlation between PET and solar radiation both in green and built-up areas, while at 4:30 p.m., PET was more related to wind velocity. During this season, the R^2 was higher in the built-up area, suggesting that green zones are less influenced by these parameters.

In winter, the same pattern was identified in the built section: better association between PET and solar radiation at 8:00 a.m. and 12:00 p.m., and with wind velocity at 4:30 p.m. On the contrary, PET in the green area presented a better correlation with wind velocity during all sessions.

PET did not vary considerably among the built-up and green areas selected for this analysis. Most probably, this was imputable to the greenery, which contributes to the UHI reduction also in its surrounding. This effect was also observed in the air temperature variation along the paths. As the areas selected for comparison are near to each other, the green area could have an effect also in the built-up area nearby and no important differences in PET were observed when comparing them. The observed cooling effect along both the pedestrian and vehicle paths suggests that the urban park, despite its smaller size, provides cooling as the larger green area traversed by the vehicle. This finding indicates that even small green spaces can mitigate urban overheating in their immediate vicinity, even if the cooling intensity and extent may be lower compared to larger areas.

5. Discussion

Results showed a higher air temperature variability registered by the vehicle in summer and in the pedestrian path during winter. Similarly to these findings, previous microclimate studies using mobile monitoring under diverse climate conditions have documented significant air temperature variability within the same city, such as in Vancouver (Tsin et al., 2016), Phoenix (Sun et al., 2009), New York City (Pioppi et al., 2022), and Singapore (Chàfer et al., 2022). These differences observed in Florence are due to the weight of solar radiation and mutual heat radiative exchange that mobile systems experienced along specific routes. The higher variability registered with the vehicle-based system can also be attributed, in a lesser extent, to other characteristics, like the longer paths and altitude variation crossed by vehicle (pedestrian path is shorter and flat). On the other hand, important research on the subject (Huang et al., 2022; Zhang and Gao, 2021) showed that the geometry of the urban built-up areas, compared with sub-urban small or large clusters, usually assessed by porosity and/or density index, has a significant impact on the thermal balance of the area. Therefore, this is closely connected to the mutual heat radiative exchange and the radiant asymmetry. Results analysis showed that the effect of radiation trapping by vertical walls becomes more significant in those areas characterized by regular geometry and medium density and porosity (i.e., the ratio of air volume within an urban canopy). These deduced interacting factors follow crucial recent literature evidence (Kinnunen et al., 2022; Lopez-Cabeza et al., 2022). Combining urban geometry with sky view factor and comparing those results obtained by vehicle with those coming from pedestrian paths allows showing how the surface temperature of the streets and multiple reflections of various surfaces/materials affects the mean radiant temperature variation. These combined effects are even more evident if the presence of intensive/extensive greenery is considered (e.g., a strong cooling effect in summer, and during winter only in the pedestrian path but in the short afternoon time also for the vehicle).

6. Limitations and future developments

While this study provides valuable insights, it is important to acknowledge a few limitations specific to the case study that may affect the generalizability of findings related to morphological impact on the observed microclimate. In this specific study, mobile monitoring campaigns were conducted solely in Florence during summer and winter, potentially capturing specific characteristics of this location, acknowledged to provide for the targeted urban complexity. However, the presented framework, combining both long-term monitoring from fixed weather stations and mobile monitoring sessions with two different transects, proved to be effective and reliable in capturing the UHI effect over the years and its intra-urban variations, identifying precise triggers for human heat stress, and assessing the specific impact of mitigation strategies, like urban greenery, on improving comfort and well-being. In particular, data from fixed weather stations, besides being long-term, are also useful to overcome data gaps of mobile monitoring sessions due to technical issues, providing records continuity. On the other hand, mobile monitoring captures the spatial variability in environmental parameters, complementing data from fixed weather stations which are usually a few within the city (just three in this case study). Furthermore, periodic repetitions of these campaigns permit assessing the impacts on the microclimate of urban dynamics, mostly related to anthropogenic activities, and verifying how these alterations affect heat stress on hyperlocal scales, which is not possible with fixed weather stations alone since they are usually placed above the urban canopy. Moreover, the combination of the wearable system with the vehicle one enables monitoring different areas, even streets that are not accessible by car, which is the case of P3 and P4 areas (Fig. 2) in the pedestrian monitoring path. Thus, the combination of different techniques provides significant benefits to urban microclimate analyses.

Scaling up the proposed framework may require additional resources due to the need for more fixed weather stations and mobile monitoring sessions. Conversely, the combined approach using both fixed and mobile monitoring potentially reduces the overall number of sensors needed compared to relying solely on either method. Consequently, targeted and strategically planned mobile monitoring sessions complementing long-term data collection can diminish the dependence on fixed stations and possibly lessen costs associated with extensive fixed-station networks.

Another limitation concerns the proposed temporal correction of the air temperature retrieved by mobile systems, based on the simplified assumption of a linear variation occurring equally across the city that could not capture some complex temporal dynamics of air temperature. Even though time correction could not be perfectly accurate, the effects of elapsed time on air temperature cannot be disregarded to evaluate the intra-urban spatial variability of this parameter, especially considering the long mobile monitoring sessions performed in this study (more than 1 h). Then, regardless of this, the time correction was performed to reduce this effect. The fact that just one fixed station (the only one on the city center) was used to evaluate the air temperature variation on time implies in assuming the same trend across the area of interest. On the other hand, the application of a consistent temporal trend to the whole path makes very time-specific changes in air temperature identified with the transects still noticeable. Therefore, this preliminary proposal demonstrated to be particularly valuable for the specific case, effectively capturing air temperature variations along the paths.

Despite these limitations, the outcomes presented reveal the robustness and potentialities of the proposed framework combining fixed and mobile monitoring techniques to evaluate urban microclimate and thermal stress, taking into account the impact of urban settings on them on a hyperlocal scale. This paper presents a real-world application of this framework through a case study, but the proposed approach is flexible and adaptable to be employed in different scenarios. In this view, as a future development to evolve the proposed framework, it is important to apply it to varied urban contexts, climate conditions, and different seasons to potentially generalize the results, as well as scale it up to verify its effectiveness when implemented in larger urban environments. This method showed to be valuable for assessing the effect of diverse urban configurations and mitigation actions, like urban

greenery, in the microclimate. Therefore, future studies could take advantage of this approach to investigate the impact of varied settings on the microclimate on a hyperlocal scale. Moreover, the results of the Kriging interpolation revealed the possibility of optimizing the mobile monitoring campaigns since acceptable errors were achieved with reduced samples. Finally, future advances of this framework involve including other thermal-related parameters like surface temperature, as well as additional domains, such as air quality, to evaluate the correlation of air pollution with the thermal issues here identified and propose mitigation actions for both domains.

7. Conclusions

Our research develops and implements a systemic approach to urban microclimate monitoring. Urban areas are indeed complex systems where buildings, energy systems, activities, various services and functions, people, and the surrounding global environment represent dynamic players of this multifunctional ecosystem in relation to natural and anthropic impacts. Findings demonstrated that the advanced prototyped monitoring technologies and methods integration (monitoring using weather fixed stations and mobile systems) could be an effective tool for studying and better managing the potential UHI mitigation. Data from networks of fixed weather stations provide main microclimate parameters variation in the long-term that could be detailed in daily and seasonal fluctuations. These profiles analysis is of utmost importance for understanding the impact of different ground uses and activities on climate. On the other hand, mobile systems address the needed spatial granularity for assessing the livability of cities and citizens' exposure to potentially temporary and harmful conditions, e.g., heat waves and local thermal emissions. The performed comparison of air temperature spatial distribution given by applying the Kriging method with the full retrieved dataset and only one-fifth of its content provides interesting insights on the amount of needed data from mobile transects to get reliable results. Future studies should explore optimizing mobile monitoring by strategically reducing the number of measured points and focusing on key driving paths, possibly setting miniaturized monitoring setups on public transportation vehicles.

The comprehensive understanding of urban microclimate guaranteed by the proposed approach can support urban planners and policymakers in proposing targeted and efficient strategies for enhancing cities' livability regarding their environmental conditions. In detail, our methodological approach can increase the effectiveness of environment conservation interventions, restoration, and adaptation to climate change; improve the interoperability of information of environmental and microclimatic interest also between local, regional and national levels; facilitate the access to national environmental information guaranteeing a more effective support for multi-level institutional reporting; strengthen and support territorial and urban planning taking into account integrated renewable sources and energy cascade management with local climate change projection; help identify priority areas for conservation, restoration, retrofitting, refurbishment, green transition, and sustainable energy design.

Finally, our findings provide critical information on the strategic spatial resolution for terrestrial and urban monitoring of most systems and ecosystems, suggesting important insights regarding adaptive governance models aimed at nature-based solutions, environmental energy sustainability, human health, green economy, and energy transition, which at the moment still lacks. Our study can be also an operative tool for providing environmental/urban thermal maps, always updatable, that can be useful for the public administration with the perspective of conservation and protection of biodiversity, green landscape, historic buildings and areas, and cultural heritage.

CRediT authorship contribution statement

Roberta Jacoby Cureau: Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft. **Carla Balocco:** Investigation, Writing – review & editing, Supervision. **Ilaria Pigliautile:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Writing – review & editing. **Cristina Piselli:** Formal analysis, Investigation, Writing – review & editing, Methodology. **Claudia Fabiani:** Data curation, Formal analysis, Investigation, Writing – review & editing. **Franco Cotana:** Supervision. **Cristina Carletti:** Formal analysis, Writing – review & editing. **Fabio Sciurpi:** Investigation, Methodology, Writing – review & editing. **Anna Laura Pisello:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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